Local-First Translation in the Browser

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Abstract

Large language models have changed how people read foreign text on the web. However, most browser translation tools still send every paragraph to a remote service, which is slow and costly. In this paper we study a local-first design in which a small translation model runs on the laptop GPU while larger models are used only when needed. We show that careful scheduling keeps the first screen under half a second and that placeholder-aware prompting preserves links, code and inline formulas. Our experiments cover news, documentation and academic papers in five languages.

1 Introduction

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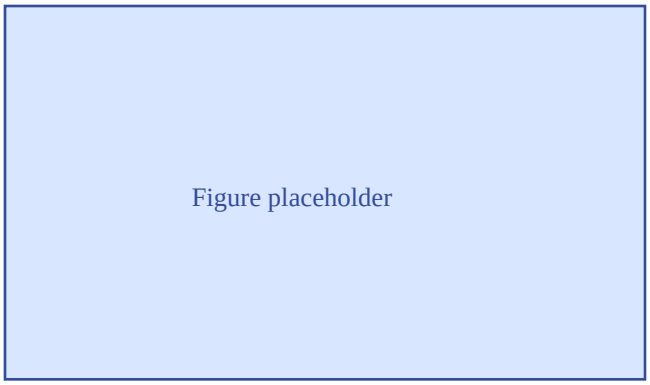

Figure 1: Latency of the first translated paragraph for each scheduling policy.

$$\alpha + \beta = \gamma \tag{1}$$

The scheduler ranks providers by latency and remaining quota, and switches before a quota is exhausted.

References

[1] A. Author. Neural machine translation. 2024.
[2] B. Author. Placeholder prompting for MT. 2025.